Keenan Heinz

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Education

University of Pennsylvania, School of Engineering and Applied Sciences Philadelphia, PA
Bachelor of Science in Engineering, Mechanical Engineering (Structures & Materials) *Aug. 2024 – May 2028*
Relevant Coursework: Dynamics, Solids, Intro to Mechatronics, Analytical Engineering Methods, Intro to Scientific Computing, Thermodynamics I, Electromagnetics

Engineering Experience

Lead Propulsion Engineer Philadelphia, PA
Penn Hyperloop *September 2025 – Present*

- Led Fusion 360 design of an A36 steel Tunnel Boring Machine propulsion structure, rated for vertical loads >2500 N and horizontal loads >5000 N.
- Performed structural FEA and hand calculations to validate integrity under worst-case loading.
- Drove actuator sourcing for a propulsion system redesign, evaluating 50+ suppliers to meet 150 kN thrust and 1+ meter stroke targets while minimizing cost.
- Currently working on next-gen TBM with ungraded propulsion structure, fluids system, and integrated steering.

Mechanical Engineering Intern Menlo Park, CA
Silicon Dreams *June 2026 – August 2026*

Automated Test Rig Fleet (Mini-Instron DAQ Systems)

- Owned mechanical design documentation: multiple 500+ line spreadsheets of design criteria, BOMs, and hand calculations.
- Validated fixture fasteners and structures for a 50 kg working load with a 2.0 yield safety factor (based on design criteria).
- Designed and built wiring and controls: STM32 MCU, precision 24-bit ADC, and 3.0 mV/V load-cell signal chain.
- Developed SPI-based DAQ software for synchronized force, displacement, and time measurements across automated test rigs.
- Applied self-taught DFM practices to prepare designs for outsourced fabrication.

Non-Contact Optical Extensometer

- Defined and validated focal length, camera position, and frame-rate limits for non-contact deformation measurement.
- Built vision-processing software for sub-pixel deformation measurement using gradient descent and camera-controller synchronization.

Coil Winding System (ITC-PLC-70)

- Designed custom bobbins and a tensioning mechanism for consistent wire feed across varying coil geometries.
- Developed HMI-integrated software to set and adjust coil-winding parameters during operation.
- Conducted test winds to validate coil-winding system scalability for future production volume.

Athletics

D1 and National Team Athlete, Rowing Philadelphia, PA
University of Pennsylvania / USRowing National Team *Aug. 2024 – Present*

- Two-time National Championship qualifier: 8th, 2026 IRA; 9th, 2025 IRA (team best points finish).
- 2nd, 2024 World Rowing Championships (St. Catharines); 5th, 2023 World Rowing Championships (Paris).

Technical Skills

CAD & Structural Analysis: Fusion 360, SolidWorks, Onshape, FEA, DraftSight, MATLAB
Controls & Instrumentation: STM32, DAQ, ADCs, load cells, stepper drivers, wiring, signal chains
Manufacturing: DFM, production engineering, MIG welding, ironworker, band saws, drill press
Programming: Python, Java, Excel, spreadsheet-based engineering calculations
Design: DaVinci Resolve, CapCut, Adobe Photoshop, Pixlr, PowerPoint, Canva
Interests: backpacking and camping, road biking, unique cars, rock music, cats